Longkun He

Ph.D. student

School of Earth and Atmospheric Sciences,

Georgia Institute of Technology, Atlanta, GA, 30332

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Education

Georgia Institute of Technology — Georgia, United States
Ph.D. student major in Atmospheric Sciences — 2024-Present
Advisor: Prof. Pengfei Liu

Peking University — Beijing, China
M.S. major in Environmental Health — 2021-2024
Advisor: Prof. Yingjun Liu

Shanghai Jiao Tong University — Shanghai, China
B.S. major in Resource and Environmental Science — 2017-2021

Selected Honors

Outstanding Graduate, Peking University — 2024
Graduate Student Scholarship, Peking University (top 10%) — 2023, 2022
Hongcai Scholarship, Peking University — 2023, 2022
Award for Scientific Research, Peking University (top 10%) — 2022
Outstanding Graduate of Shanghai, Shanghai Jiao Tong University — 2021
Distinguished Thesis, Shanghai Jiao Tong University (top 5%) — 2021
First Prize Scholarship, Shanghai Jiao Tong University (top 10%) — 2019
National Scholarship, Shanghai Jiao Tong University (top 1%) — 2018

Teaching Experience & University Service

Teaching Assistant — Georgia, United States
EAS 1600, School of Earth and Atmospheric Sciences, Georgia Institute of Technology — 2024-2025

- Prepared lectures and lab experiments for undergraduates.
- Graded course assessments to ensure students understood material and stayed on track.

International Teaching Assistant Liaison — Georgia, United States
Center for Teaching and Learning, Georgia Institute of Technology — 2025-present

- Served as a vital bridge between the Center for Teaching and Learning and International Teaching Assistants (ITAs) within the department.
- Advocated for ITA needs by recognizing their unique challenges and actively connecting new and current teaching assistants with tailored CTL resources and workshop series.
- Fostered a more inclusive academic environment by interviewing faculty and promoting initiatives that support the professional development of ITAs.

Professional Service

Reviewer

2 in *Atmospheric Environment: X* 2025
2 in *Journal of Aerosol Science* 2025
2 in *Aerosol and Air Quality Research* 2025

Research Experience

Field Study

Gas-particle partitioning of SVOCs during wintertime Peking University, 2021-2024

- Measured concentrations of gas-phase and particle-phase SVOCs by semivolatile thermal desorption aerosol gas chromatography (SV-TAG)
- Investigated effects of temperature and humidity on gas-particle partitioning of SVOCs

Hygroscopic and optical properties of ambient particles Georgia Institute of Technology, 2024-present

- Develop the humidified nephelometer system for high-resolution measurements of aerosol optical properties
- Investigate the hygroscopic and optical properties of ambient particles

Lab Experiment

Temperature dependence of wall loss of SVOCs in a Teflon chamber Peking University, 2022-2023

- Conducted wall-loss experiments of *n*-alkanes in a Teflon chamber for a wide temperature range
- Observed enhanced wall loss for less volatile alkanes and lower temperatures

Model Simulation

Vapor−wall mass transport of SVOCs in a Teflon chamber Peking University, 2022-2023

- Employed two-layer kinetic model to analyze wall-loss data
- Investigated two kinds of processes in vapor-wall interaction

Dynamics of SVOCs in an office Peking University, 2021-2022

- Built a kinetic model to simulate SVOC dynamics in the indoor air
- Explained the observed fast temperature response and indiscernible ventilation effect on SVOCs dynamics

Predicting ALWC using machine learning algorithm Georgia Institute of Technology, 2025-present

- Use pseudo aerosol model to generate training datasets based on physics
- Predict ALWC by ML model on real-world datasets

Publications

- Peer-reviewed journal articles: 9
- Conference Abstract: 5

https://scholar.google.com/citations?user=cuvYdUwAAAAJ

Journal Articles (* Co-first Author):

1. Liu, W.*, **He, L.***, Li, Y., Li, P., Chen, S., Chen, J., Liu, Y., and Kuwata, M.: Prolonged atmospheric chemical lifetime of unsaturated fatty acids from cooking sources observed in Beijing during wintertime, *Journal of Geophysical Research: Atmospheres*, 130, e2025JD044172, https://doi.org/10.1029/2025JD044172, 2025.
2. **He, L.**, Liu, W., Li, Y., Wang, J., Kuwata, M., and Liu, Y.: Wall loss of semi-volatile organic compounds in a Teflon bag chamber for the temperature range of 262–298 K: mechanistic insight on temperature dependence,

Atmospheric Measurement Techniques, 17, 755–764, https://doi.org/10.5194/amt-17-755-2024, 2024.

3. Zhou, Y., Wang, Z., Wu, C., Li, J., **He, L.**, Zhang, B., Guo, J., and Zhang, J.: Inhibition of imidacloprid photolysis by particulate matter: Experimental and computational evidence, *Environmental Science & Technology*, 60 (11), 8681-8691, https://doi.org/10.1021/acs.est.5c18139, 2026.
4. Li, Y., Zhao, Y., Shen, X., Xue, Y., **He, L.**, Xu, S., Yang, L., Yi, J., Zhang, J., Fan, G., and Deng, Q.: Per- and polyfluoroalkyl substances on the surface of children's feeding smocks: High-throughput characterization and implications for exposure assessment, *Environmental Chemistry and Ecotoxicology*, 7, 1715-1722, https://doi.org/10.1016/j.enceco.2025.08.001, 2025.
5. Zhou, Y., **He, L.**, Tan, J., Zhou, J., and Liu, Y.: Rapid quantitative analysis of semi-volatile organic compounds in indoor surface film using direct analysis in real time mass spectrometry: a case study on phthalates, *Atmospheric Measurement Techniques*, 17, 6415–6423, https://doi.org/10.5194/amt-17-6415-2024, 2024.
6. Liu, W., **He, L.**, Liu, Y., Liao, K., Chen, Q., and Kuwata, M.: Suppressed atmospheric chemical aging of cooking organic aerosol particles in wintertime conditions, *Atmospheric Chemistry and Physics*, 24, 5625-5636, https://doi.org/10.5194/acp-24-5625-2024, 2024.
7. Liu, W., Liao, K., Chen, Q., **He, L.**, Liu, Y., and Kuwata, M.: Existence of Crystalline Ammonium Sulfate Nuclei Affects Chemical Reactivity of Oleic Acid Particles Through Heterogeneous Nucleation, *Journal of Geophysical Research: Atmospheres*, 128, https://doi.org/10.1029/2023jd038675, 2023.
8. Li, Y., **He, L.**, Xie, D., Zhao, A., Wang, L., Kreisberg, N., Jayne, J., and Liu, Y.: Strong temperature influence and indiscernible ventilation effect on dynamics of some semivolatile organic compounds in the indoor air of an office, *Environment International*, 165, https://doi.org/10.1016/j.envint.2022.107305, 2022.
9. Li, Y., Xie, D., **He, L.**, Zhao, A., Wang, L., Kreisberg, N., Jayne, J., and Liu, Y.: Dynamics of di-2-ethylhexyl phthalate (DEHP) in the indoor air of an office, *Building and Environment*, 223, https://doi.org/10.1016/j.buildenv.2022.109446, 2022.

Conference Abstract:

1. Barut, C., Hagan, D., Liu, R., Jeon, S., **He, L.**, Liu, P., Baliaka, H. D., Ward, R., Seinfeld, J., Flagan, R., Ries, B., Bowey, E., Pena, G., Cui, Y., Bahreini, R., Rogers, M., Joo, T., Hass-Mitchell, T., Gentner, D., Presto, A., Yun, S., Day, D. A., Jimenez, J.-L., and Ng, N. L.: Multi-site evaluation and physical correction of QuantAQ PM2.5 measurements using ASCENT research grade measurements, *44th Annual Conference of the American Association for Aerosol Research*, 2026.
2. Gwak, S., Jeon, S., Ivančič, M., Lavrič, G., Liu, R., **He, L.**, Liu, P., Gregorič, A., Rigler, M., Yadav, V., and Ng, N. L.: Year-long online quantification and source apportionment of brown carbon in Atlanta, Georgia, *44th Annual Conference of the American Association for Aerosol Research*, 2026.
3. Rigler, M., Ivančič, M., Ježek Brecelj, I., Lavrič, G., Alföldy, B., Liu, R., Jeon, S., **He, L.**, Liu, P., Ng, N. L., and Gregorič, A.: High-time-resolution measurements of total and black carbon and the determination of primary and secondary organic carbon at the ASCENT South DeKalb (Atlanta, GA) measurement site, *43rd Annual Conference of the American Association for Aerosol Research*, 2025
4. Liu, W., Kuwata, M., **He, L.**, Li, Y., Liu, Y., Liao, K., Chen, Q., and Chen, S.: Suppressed atmospheric chemical aging of cooking organic aerosol particles in wintertime conditions: Evidence from laboratory and ambient measurements, *AGU Fall Meeting Abstracts*, 2023.
5. Liu, Y., Li, Y., **He, L.**, Zhao, A., Wang, L., Kreisberg, N. M., and Jayne, J. T.: Influence of environmental factors on indoor SVOC dynamics: Insights from online observations in an office, *AGU Fall Meeting Abstracts*, 2022.